

The Cobordism Functor, the Theta Register, and Root Systems

Problems

Problems for `docs/design/cobordism_functor_reading.tex`, in the order of its sections; theorem, proposition and section numbers refer to that document and are quoted with their names. Problems marked $(\star)$ are the spine of the framework; the others are context and practice. Hints follow each section; complete worked solutions are in the companion `cobordism_functor_solutions.tex`. Code exercises use the package `tessera` and `examples/cobordism/qubit_animation.py`; their records land under `~/cobordism-runs/`.

The primer

1 Cochains, the Laplacian and the harmonic space (Section 2)

- P1.** $(\star)$ Prove the Leibniz rule $d(\alpha \cup \beta) = d\alpha \cup \beta + (-1)^{|\alpha|} \alpha \cup d\beta$ for simplicial cochains, and deduce the orthogonality half of Theorem 3.1 (mutual annihilators).
- P2.** $(\star)$ The minimal Δ -complex torus has one vertex v , three loop edges a, b, c and two triangles U, L with $\partial U = a + b - c$, $\partial L = -(a + b - c)$. Write d_0 and d_1 , verify $d_1 d_0 = 0$, and compute the Betti numbers. Now give the faces masses $M_2 = \text{diag}(m_U, m_L)$ and the edges any invertible diagonal mass M_1 . Show that when $m_L = -m_U$ the Hodge Laplacian L_1 vanishes identically, so $\dim \ker L_1 = 3 \neq b_1$. Which of the certificates of Section 26 (closedness on images, coclosedness on the chain frame, an independent kernel of the same span) detects the failure, and why does Theorem 2.2 (discrete Hodge) not apply?
- P3.** Prove Theorem 2.4 (periods do not move) and explain what the verifier's checks J1–J3 measure: under a jitter of the edge lengths the monodromy is unchanged, the harmonic representative of a class moves, and the difference of the two representatives is exactly a coboundary.

Hints. P1: check on a single simplex, then extend linearly. P2: $d_0 = 0$ because every edge is a loop; $d_1 \alpha = (x, -x)$ with $x = \alpha_a + \alpha_b - \alpha_c$.

2 Restriction and the monodromy (Sections 3–4)

- P4.** $(\star)$ Write the long exact sequence of $(W, \partial W)$ in degrees 0, 1, 2 for $W = T^2 \times [0, 1]$ and, using Lefschetz duality, verify $\text{rank}(H^1(W) \rightarrow H^1(\partial W)) = 2 = \frac{1}{2} b_1(\partial W)$.
- P5.** Describe $L_W \subset H^1(M_0) \oplus H^1(M_1) \cong \mathbb{C}^4$ for the product collar concretely, and show it is isotropic for $\Omega_{\partial} = J \oplus (-J)$.

- P6.** (★) For 2×2 matrices, $M^\top JM = \det(M) J$. Show this fails for 4×4 by computing $M^\top (J \oplus J) M$ for $M = \sigma \oplus \sigma$, σ the swap: $\det M = +1$ yet the form is reversed. Conclude that the orientation sign of Remark 3.3 must be declared per torus, not derived from $\det M$.
- P7.** Show $M = \varphi^*$ for the collar whose far end is relabelled by a simplicial automorphism φ , and compute φ^* for the swap and for the Dehn twist $(x, y) \mapsto (x + y, y)$. Which is simplicial on the square grid with one diagonal, and why not the other? Then explain how the layered flip passes of Section 35.2 realize the twist anyway: what is the flipped grid, and why does one tetrahedron per flipped edge suffice?
- P8.** (★) The code reports, under a 75% complex jitter of every squared length, that M moves by 1.6×10^{-14} while the harmonic representative ZP_A^{-1} moves by 0.7 with the difference exactly in $\text{im } d_0$. Name the theorem each number measures, and say what a nonzero non-exact part would have meant.
- P9.** Theorem 4.2 needs an integral hypothesis. Take $P_A = \text{diag}(1, 2)$ and $P_B = \text{diag}(2, 1)$: show M is symplectic but not integral, and explain why invertibility of P_A over $\mathbb{C}$ cannot replace the lattice hypothesis. Then take the connected sum of a product collar with a solid torus $S^1 \times D^2$ (along a ball). Show that $H^1(W)$ gains a class with zero periods on both boundary tori, that P_A becomes 2×3 , and that M is nevertheless unique (compare the identifiability proposition of Section 26).

Hints. P4: $H^1(W, \partial W) \cong H_2(W) \cong H_2(T^2) = \mathbb{Z}$. P6: $J\sigma = \text{diag}(1, -1)$. P8: Theorem 2.4 and Lemma 2.1.

3 Choi vectors and composition (Sections 5–6)

- P10.** (★) Prove that a subspace $L \subset A \oplus B$ is the graph of a linear map iff $L \cap (0 \oplus B) = 0$ and L projects onto A , and that the map is then unique. Identify $\dim L$.
- P11.** Write $|M\rangle = \sum_i M e_i \otimes e_i^*$ for $M = I$ and $M = \sigma$ in the basis $e_i \otimes e_j^*$ of $\mathbb{C}^2 \otimes (\mathbb{C}^2)^*$. Which is the maximally entangled Bell vector, and what does that say about the product collar?
- P12.** (★) Carry out the Mayer–Vietoris argument of Theorem 6.1 for two product collars glued end to end, and show the composite is a product collar with $M = I \cdot I$.
- P13.** (★) Prove Theorem 6.1 in general: for $W = W_1 \cup_{M_1} W_2$ the boundary relation is the composite of relations, $L_W = L_{W_2} \circ L_{W_1}$, where $L_2 \circ L_1 = \{(x_0, x_2) : \exists x_1, (x_0, x_1) \in L_1, (x_1, x_2) \in L_2\}$. Where does exactness of Mayer–Vietoris enter, and why do invisible classes (the image of the connecting map from $H^0(M_1)$) not spoil the statement?
- P14.** The verifier’s check C1 compares $M(2L, \text{none})$ with $M(L, \text{swap})^2$ on *separately seeded* collars. Explain why this tests the group law but not a seam. Then explain what the flip passes (F4, T9) add — $T_A T_B$ and $T_B T_A$ stacked on one host read different matrices — and what they still do not test.

Hints. P10: the projection to A restricted to L is injective iff the first condition holds. P11: $\sum_i e_i \otimes e_i^*$. P13: both inclusions; the connecting map changes the extension, not its restrictions. P14: $\{\pm I, \pm \sigma\}$ is abelian; flip passes live on one complex.

4 What the metric controls (Section 7)

- P15.** Using Remark 7.1, show that $\Phi^\top G \Phi$ changes under a vertex gauge but $\tilde{\Phi}^\top G \Phi$ does not, and explain why the two agree when every phase is zero. (The code measured 0.76 against 1.3×10^{-15} .)
- P16.** Proposition 7.2 says the pinned-boundary defect $D = \|(M^\vee)^\top G_B M - G_A\|_F / \|G_A\|_F$ is invariant under interior metric changes. Prove it, say why a relaxation of the bulk therefore cannot lower it, and explain why $D = 0$ would still not certify positive quantum unitarity of M .

Hints. P15: at zero phases $\tilde{\Phi} = ZB^{-\top}$ with $B = \Phi^\top Z$.

5 Weights, phase space and quantization (Sections 8–9)

- P17.** (★) For a torus group $T = \mathfrak{h}/\Lambda$ show that the characters are $e^{2\pi i \langle \mu, \cdot \rangle}$ with $\mu \in \Lambda^*$, that the weights of $V \otimes V'$ are the sums of weights, and that the weight lattice of $T_1 \times T_2$ is $\Lambda_1^* \oplus \Lambda_2^*$. Explain why “all characters” is an infinite-dimensional space and what theta quantization adds to obtain a finite register.
- P18.** (★) Let ϕ be a real closed 1-cochain on a genus- g surface with normalized periods a_i, b_i (Section 9). Show that gauge transformations do not change (a, b) , that large gauge transformations shift them by integers, and that $J(\Sigma) = H^1(\Sigma; \mathbb{R})/H^1(\Sigma; \mathbb{Z})$ is a real torus of dimension $2g$ on which $\sum_i da_i \wedge db_i$ has integral periods equal to the intersection form. With $z = b - \Omega a$, show that the integer shifts of (a, b) are exactly the lattice $\mathbb{Z}^g + \Omega \mathbb{Z}^g$ acting on z , so that z is a coordinate on the complex torus $\mathbb{C}^g/(\mathbb{Z}^g + \Omega \mathbb{Z}^g)$.
- P19.** On the torus with one vertex, two edges and one face, the differentials of the rank-one local system with holonomies u, v are $d_0 = \begin{pmatrix} u-1 \\ v-1 \end{pmatrix}$ and $d_1 = \begin{pmatrix} -(v-1) & u-1 \end{pmatrix}$. Verify $d_1 d_0 = 0$ and show that for $(u, v) \neq (1, 1)$ every cohomology group vanishes. Why does this show that the state-selection phases must be stored separately from the neutral Hodge assembly?

Hints. P18: $a \mapsto a + e_i$ sends $z \mapsto z - \Omega e_i$.

6 Theta functions and the tensor register (Sections 10–11)

- P20.** (★) Prove the periodicity $\theta_j(z+1) = \theta_j(z)$ and the quasi-periodicity of Section 10 directly from the sum.
- P21.** (★) Show that $\|\theta_j\|(z) = |\theta_j(z; \tau)| e^{-\pi k(\text{Im } z)^2 / \text{Im } \tau}$ is invariant under both translations. (This is the quantity the record’s figure draws; $|\theta_j|$ alone is dominated by the growth factor.)
- P22.** Using the argument principle around a fundamental parallelogram and the quasi-periodicity factor, show each θ_j has exactly k zeros there. Locate them for $k = 2, j = 0, 1$.
- P23.** (★) Verify the clock and shift formulas of Section 10 and the commutation $ZX = e^{2\pi i/k} XZ$.
- P24.** Prove Proposition 11.1 (factorization), and explain why the same argument fails for a non-diagonal Ω .

- P25.** (★) Derive the normalization of Theorem 11.2 for $g = 1$: with $z = x + \tau y$ show that $\sqrt{2k}\sqrt{\text{Im } \tau} \int_0^1 \int_0^1 \overline{\theta_j} \theta_{j'} e^{-2\pi k y^2 \text{Im } \tau} dx dy = \delta_{jj'}$. Explain why the weight $e^{-2\pi k y^2 \text{Im } \tau}$ is the square of the factor in P21, and how the argument generalizes to g variables.

Hints. P21: compute how each factor changes under $z \mapsto z + \tau$. P22: the winding of θ_j around the boundary of the parallelogram is the total change of arg; the two vertical sides cancel and the two horizontal sides differ by the factor $e^{-2\pi i k z}$. P25: integrate in x first; the surviving terms combine into one Gaussian in $n + j/k + y$.

7 The geometric operator algebra (Section 12)

- P26.** (★) With $X_r|j\rangle = |j + \epsilon_r\rangle$, $Z_r|j\rangle = \omega^{j_r}|j\rangle$ and $D_{p,q} = X^p Z^q$, prove $Z^q X^{p'} = \omega^{q \cdot p'} X^{p'} Z^q$ and deduce the product and involution laws of the reduced algebra, $D_{p,q} D_{p',q'} = \omega^{q \cdot p'} D_{p+p', q+q'}$ and $D_{p,q}^\dagger = \omega^{p \cdot q} D_{-p, -q}$. Check associativity from the scalar cocycle.
- P27.** (★) Show $\text{Tr}(D_{p,q}^\dagger D_{p',q'}) = d \delta_{pp'} \delta_{qq'}$ with $d = k^g$, conclude that the d^2 operators $D_{p,q}$ form an orthogonal basis of $\text{End}(\mathcal{Q}_k)$ with inverse formula $T = \frac{1}{d} \sum_{p,q} \text{Tr}(D_{p,q}^\dagger T) D_{p,q}$, and derive the twisted convolution $(a \star b)_{r,s} = \sum_{p,q} a_{p,q} b_{r-p, s-q} \omega^{q \cdot (r-p)}$. Verify it by hand for $k = 2$, $g = 1$ on $a = e_{1,0}(X)$, $b = e_{0,1}(Z)$: compute $a \star b$ and $b \star a$ and compare with XZ and ZX .
- P28.** Corollary 12.3 identifies states with positive symbols of geometric trace one, $\text{tr}_G(a) = d a_{0,0}$. For $k = 2$, $g = 1$ find the symbol of $|0\rangle\langle 0|$ and of the effect $e = |+\rangle\langle +|$, check tr_G of the state is 1, and compute the Born probability $\text{tr}_G(ae)$ without forming any matrix.
- P29.** Prove Theorem 12.4 in coordinates: $D_{(p_1, p_2), (q_1, q_2)} = D_{p_1, q_1} \otimes D_{p_2, q_2}$ in the factorized basis, the vectors $|D_{p,q}\rangle / \sqrt{d}$ are orthonormal, and contracting the dual intermediate factor of $|T_2\rangle \otimes |T_1\rangle$ gives $|T_2 T_1\rangle$. Do the last step with matrix units E_{ij} and explain why intermediate renormalization would lose information.

Hints. P26: $Z_r X_s = \omega^{\delta_{rs}} X_s Z_r$ on basis vectors. P27: a nonzero shift has zero diagonal; a nonconstant character sums to zero. P28: $|0\rangle\langle 0| = (I + Z)/2$, $|+\rangle\langle +| = (I + X)/2$.

8 The modular action and the Weil representation (Section 13)

- P30.** (★) Derive $\rho_k(T) = \text{diag}(e^{\pi i j^2 / k})$ for k even, and show what goes wrong for k odd.
- P31.** (★) Derive $\rho_k(S)$ by Poisson summation: apply $\sum_n f(n) = \sum_m \hat{f}(m)$ to the Gaussian in the definition of $\theta_j(z/\tau; -1/\tau)$. Recover the Hadamard gate at $k = 2$. Show that the identity is exact with $j(S, \tau) = (-i\tau)^{1/2}$ and no further constant, and that at $k = 4$ the sign in $e^{-2\pi i j l / k}$ is forced (the conjugate matrix fails).
- P32.** Show $\rho_k(S)^2$ is the parity operator $j \mapsto -j$ (up to a phase), and $(\rho_k(S) \rho_k(T))^3 \propto \rho_k(S)^2$, matching $S^2 = -I$ and $(ST)^3 = S^2$ in $\text{SL}(2, \mathbb{Z})$.
- P33.** (★) Verify the shear formula: for $\Omega \mapsto \Omega + B$, θ_j picks up $e^{\pi i j^T B j / k}$ when k is even and B symmetric integer. Compute it for $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $k = 2$, and identify controlled- Z .
- P34.** (★) Prove Proposition 13.3 (products), and compute the operator Schmidt rank of $\rho_2(\text{shear})$ across the two qubits.

- P35.** Explain why $\rho_k(\mathbf{M})$ cannot depend on τ , given that the transformation law holds for all τ and the functions $\theta_l(\cdot; \tau)$ are linearly independent. (The code measures the fitted matrices at two moduli to agree to 9×10^{-16} , check T7.)
- P36.** (★) Prove Proposition 13.1 (the coordinate bridge). A period map $M_{\text{per}} = \begin{pmatrix} P & Q \\ R & S \end{pmatrix}$ acts on (a, b) . Show that the new normalized period matrix is $\Omega' = (R + S\Omega)(P + Q\Omega)^{-1}$, that $z' = b' - \Omega'a' = (P + Q\Omega)^{-\top} z$, and that the standard Siegel matrix acting by $\Omega' = (A\Omega + B)(C\Omega + D)^{-1}$ is $\mathbf{M} = \begin{pmatrix} S & R \\ Q & P \end{pmatrix}$. Specialize to genus one and compare with the measured far modulus of the flipped torus (F3), $\tau' = (M_{21} + M_{22}\tau)/(M_{11} + M_{12}\tau)$.
- P37.** Prove Proposition 13.4 (compatible geometric transport is unitary): the clock operators have a common eigenvector whose shift orbit spans an irreducible representation of the finite Heisenberg group; a compatible relabelling gives a second such representation, and the intertwiner is unique up to a scalar, which adjoint-preservation makes a phase.

Hints. P30: $e^{\pi i k n^2} = (-1)^{kn}$. P31: the Fourier transform of $e^{\pi i k(n+j/k)^2 \tau}$ in n is again Gaussian with $\tau \mapsto -1/\tau$; the characteristic j becomes a phase $e^{-2\pi i j m/k}$. P36: $\Omega'(P + Q\Omega) = R + S\Omega$ and symplecticity give $S - \Omega'Q = (P + Q\Omega)^{-\top}$. P37: Schur's lemma for the algebra spanned by the $D_{p,q}$, which is all of $\text{End}(\mathcal{Q}_k)$ (P27).

9 Orientation reversal and coherent states (Sections 14–15)

- P38.** Show $\overline{\theta_j(z; \Omega)} = \theta_j(-\bar{z}; -\bar{\Omega})$ and hence that orientation reversal acts antiunitarily. For the torus period swap $M_{\text{per}} = \sigma$, form $N = M_{\text{per}} R_{\text{per}}$ with $R_{\text{per}} = \text{diag}(1, -1)$, check that N is symplectic, identify N and its Siegel matrix $\mathbf{N}$, and write the level-two action as a matrix composed with conjugation.
- P39.** (★) Prove Proposition 15.2: if $F_{\Omega'}(z') = f \rho F_{\Omega}(z)$ with ρ unitary and $f \neq 0$, then the normalized conjugate evaluation vectors satisfy $c_{\Omega'}(z') = (\bar{f}/|f|) \bar{\rho} c_{\Omega}(z)$. Explain why fitting ρ on value columns and applying it unconjugated to coherent states is wrong, and why the two conventions are equally complete.
- P40.** (★) Prove Theorem 15.3: for an elliptic curve E and a degree-two line bundle L^2 , the two basis sections have no common zero, the coherent-state map $E \rightarrow \mathbb{CP}^1$ is onto and of degree two, and conjugation preserves surjectivity. Then show that at fixed polarization a single elliptic curve cannot cover all putrit rays.
- P41.** Show that for $\Omega = \text{diag}(\tau_1, \tau_2)$ the coherent state factorizes, $c_{\Omega}(z) = c_{\tau_1}(z_1) \otimes c_{\tau_2}(z_2)$, so its entanglement entropy in the factorized basis is zero, and explain why a nonzero Ω_{12} can make it nonzero even though the monodromy of the tube host is the identity (G2–G4).

Hints. P40: Riemann–Roch on genus one: $h^0(D) = \deg D$ for $\deg D > 0$.

10 Seams, factorization and the Clifford limit (Sections 16–17)

- P42.** (★) Write out Mayer–Vietoris for the sphere join and prove Proposition 16.1. The code measured the off-block norm of the four-torus monodromy at 5×10^{-15} (check D1); which map in the sequence is that number testing?

- P43.** (★) Do the same for a tube identifying a cycle a_1 on one torus with a cycle a_2 on the other, with overlap retracting to S^1 . Show the restriction subspace satisfies $p_1(a_1) = p_2(a_2)$, write the simplest monodromy consistent with it, and compute its ρ_2 and Schmidt rank.
- P44.** Why does attaching a 1-handle (a solid tube glued along two disks) *not* create a shared class, while gluing along an annulus does? Reconcile this with the tube host of Section 35.2: its monodromy is the identity (G2, “the tube alone couples no cycle”) while the far surface’s period matrix has $\Omega_{12} \neq 0$ (G3). Which of the two is topological and which is metric?
- P45.** (★) Compute $CZ|++\rangle$ and its reduced state on the first qubit. Show that SWAP maps every product state to a product state although its operator Schmidt rank is four. Define the product-input entangling witness used in E5 and explain why operator Schmidt rank alone (Proposition 13.3) is not it.
- P46.** (★) Show that the Weil representation normalizes the Heisenberg–Weyl group: $\rho(S)X\rho(S)^{-1} \propto Z$. Why does this make its image the Clifford group, and why does Gottesman–Knill then say the theta register alone is classically simulable?
- P47.** Reduction modulo 2 maps $\mathrm{Sp}(4, \mathbb{Z}) \rightarrow \mathrm{Sp}(4, \mathbb{F}_2) \cong S_6$, and the two-qubit local Cliffords with the swap map onto a subgroup isomorphic to $S_3 \wr \mathbb{Z}/2$. Use orders to show that no gate whose symplectic image has order five modulo 2 is a local Clifford or a swap times one, and verify the hypotheses for the decagon monodromy $M = \begin{pmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \end{pmatrix}$: $\det M = 1$, $M^\top J M = J$, $M^5 = I$, and $M \bmod 2$ has order five.

Hints. P43: the kernel of $H^1(W_1) \oplus H^1(W_2) \rightarrow H^1(S^1)$. P44: $W_1 \cup \text{handle} \cup W_2 \simeq W_1 \vee W_2$. P46: X and Z are translations by τ/k and $1/k$; S exchanges the roles of 1 and τ . P47: $|S_3 \wr \mathbb{Z}/2| = 72$.

11 From a Lie algebra to its root system; the Weyl group (Sections 18–19)

- P48.** (★) From $[h, e] = 2e$, $[h, f] = -2f$, $[e, f] = h$ alone, show that a finite-dimensional irreducible $\mathfrak{sl}(2)$ -module with highest h -eigenvalue λ has eigenvalues $\lambda, \lambda - 2, \dots, -\lambda$, each once, so that $\lambda \in \mathbb{Z}_{\geq 0}$ and the dimension is $\lambda + 1$. Identify $\alpha, \alpha^\vee, \omega, P, Q, P/Q$ and W in this notation, and check $|\omega|^2 = \frac{1}{2}$ when $|\alpha|^2 = 2$.
- P49.** (★) For $\mathfrak{sl}(3, \mathbb{C})$ with $\mathfrak{h}$ the traceless diagonal matrices, show that the roots are $\varepsilon_i - \varepsilon_j$ with root vectors E_{ij} ; take $\alpha_1 = \varepsilon_1 - \varepsilon_2$, $\alpha_2 = \varepsilon_2 - \varepsilon_3$ and compute the Cartan matrix, the angle between the simple roots, the highest root, ω_1, ω_2 in terms of $\alpha_1, \alpha_2, P/Q, \rho, h^\vee$, and W as a group of permutations. Draw the six roots and the weight lattice and mark P^+ .
- P50.** Write the Cartan matrix and the Gram matrix of the simple roots for $A_1 \times A_1, A_2, B_2, G_2$ with long roots of length $\sqrt{2}$. Show that the root lattices of $A_1 \times A_1$ and B_2 are both square and those of A_2 and G_2 both hexagonal (up to scale and rotation), and that $W(A_1 \times A_1) \subset W(B_2)$ and $W(A_2) \subset W(G_2)$ have index two, the missing generator being the reflection in a short root.
- P51.** (★) Prove Proposition 18.3: a Euclidean lattice generated by a triangle with side lengths $\sqrt{2}, \sqrt{2}, \sqrt{6}$ has Gram matrix $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$, its six shortest vectors form A_2 , their reflections preserve the lattice and generate S_3 , and the dual lattice is $P = A^{-1}\mathbb{Z}^2$ with $|P/Q| = 3$. Which

of these facts uses only scalar length data, and what would a boundary triangulation have to satisfy for its harmonic mass Gram to equal A ?

- P52.** (★) From the Weyl character formula (Theorem 19.1) derive $\text{ch } V_\lambda = x^\lambda + x^{\lambda-2} + \cdots + x^{-\lambda}$ for $\mathfrak{su}(2)$, and from the dimension formula derive $\dim V_{(a,b)} = \frac{1}{2}(a+1)(b+1)(a+b+2)$ for $\mathfrak{su}(3)$; check $(1,0) \mapsto 3$, $(1,1) \mapsto 8$, $(3,0) \mapsto 10$, $(2,2) \mapsto 27$.
- P53.** (★) Prove Proposition 19.2 (Brauer–Klimyk) from the character formula. Apply it to $V_1 \otimes V_\lambda$ and to $V_2 \otimes V_0$ for $\mathfrak{su}(2)$, and to $\mathbf{3} \otimes \mathbf{3}$ and $\mathbf{3} \otimes \bar{\mathbf{3}}$ for $\mathfrak{su}(3)$, exhibiting a term a wall kills and a term a reflection folds with a sign.
- P54.** Verify the Weyl denominator identity $\sum_w \varepsilon(w) e^{w\rho} = e^\rho \prod_{\alpha > 0} (1 - e^{-\alpha})$ by direct expansion for A_1 and for A_2 (six terms against eight, two of which cancel), identifying each surviving term with an element of W and its sign.

Hints. P48: e raises and f lowers the h -eigenvalue by 2; compute $ef^n v$ on a highest-weight vector v by induction and find where the string $f^n v$ must terminate. P49: $\langle \varepsilon_i, \varepsilon_j \rangle = \delta_{ij} - \frac{1}{3}$ is the trace form restricted to the traceless part; it gives $|\varepsilon_i - \varepsilon_j|^2 = 2$. P51: cosine law: $\alpha_1 \cdot \alpha_2 = (|\alpha_1|^2 + |\alpha_2|^2 - |\alpha_1 - \alpha_2|^2)/2$. P52: for $\mathfrak{su}(3)$ the positive coroots are $\alpha_1, \alpha_2, \theta$ and $\langle \rho, \cdot \rangle = 1, 1, 2$ on them. P53: the weights of $\mathbf{3}$ are $\omega_1, \omega_1 - \alpha_1 = \omega_2 - \omega_1, \omega_1 - \alpha_1 - \alpha_2 = -\omega_2$; those of $\bar{\mathbf{3}}$ are their negatives.

12 Affine algebras and theta functions (Section 20)

- P55.** (★) With t_γ acting at level K by $\mu \mapsto \mu + K\gamma$ on the finite part, show $w t_\gamma w^{-1} = t_{w\gamma}$, so that W and $t(Q^\vee)$ generate $W \ltimes t(Q^\vee)$; show that $s_0 = t_{\theta^\vee} s_\theta$ is the reflection in the hyperplane $\langle \mu, \theta^\vee \rangle = K$; and for $\mathfrak{su}(2)$ at $K = k + 2$ compute $s_0(m\omega) = (2(k+2) - m)\omega$. Conclude that the fundamental alcove is a simplex and that its interior lattice points at $K = k + h^\vee$ are exactly $P_k^+ + \rho$.
- P56.** (★) Using $t_\gamma(\hat{\mu}) = \hat{\mu} + K\gamma - (\langle \mu, \gamma \rangle + \frac{K}{2}|\gamma|^2)\delta$ and $e^{-\delta} = q$, show that $\sum_{\gamma \in Q^\vee} e^{t_\gamma(\hat{\mu})} = q^{-|\mu|^2/2K} \Theta_\mu^{(K)}$ with $\Theta_\mu^{(K)}$ as in Theorem 20.2, and conclude that the Weyl–Kac numerator is $\sum_{w \in W} \varepsilon(w) \Theta_{w(\lambda+\rho)}^{(K)}$ up to a power of q (why is the power the same for every w , and why is $\varepsilon(t_\gamma) = +1$?). State where in the computation the non-abelian structure enters and where the lattice does.
- P57.** (★) From $h_\lambda = C_2(\lambda)/2(k + h^\vee)$, $c = k \dim \mathfrak{g}/(k + h^\vee)$ and the strange formula $|\rho|^2/2h^\vee = \dim \mathfrak{g}/24$, derive $T_\lambda = \exp 2\pi i [|\lambda + \rho|^2/2(k + h^\vee) - \dim \mathfrak{g}/24]$. Specialize to $\mathfrak{su}(2)_k$; compute c for $\mathfrak{su}(2)_1, \mathfrak{su}(2)_2, \mathfrak{su}(3)_1$ and h_λ for the $\mathfrak{su}(2)_1$ doublet, and compare T with $\rho_2(T)$ of the abelian register.
- P58.** (★) (Frenkel–Kac.) Show that the two coset theta functions $\Theta_{Q+j\omega}$ of A_1 ($Q = \sqrt{2}\mathbb{Z}$) are the level-two functions $\theta_j^{(2)}$ of Definition 10.1 (with z rescaled), and deduce S and T of $\mathfrak{su}(2)_1$ from the abelian register and the multiplier system of η ($\eta(\tau+1) = e^{\pi i/12}\eta(\tau)$, $\eta(-1/\tau) = \sqrt{-i\tau}\eta(\tau)$). Then do $\mathfrak{su}(3)_1$: the three cosets of the hexagonal lattice, S the Fourier transform of $\mathbb{Z}/3$, $T \propto \text{diag}(1, e^{2\pi i/3}, e^{2\pi i/3})$. Explain why the even lattice A_2 has a diagonal T while the rank-one register at odd level (P30) does not, and compare the $\mathfrak{su}(2)_1$ fusion rule with the Heisenberg–Weyl composition of characteristics.

Hints. P55: $\theta^\vee = \theta$ since θ is long; the reflection in the affine hyperplane $\langle \mu, \theta \rangle = K$ along θ is $\mu \mapsto \mu - (\langle \mu, \theta \rangle - K)\theta$. P56: $\frac{K}{2}|\frac{\mu}{K} + \gamma|^2 = \frac{|\mu|^2}{2K} + \langle \mu, \gamma \rangle + \frac{K}{2}|\gamma|^2$, and $t_{\theta^\vee} = s_0 s_\theta$. P57: for $\mathfrak{su}(2)$, $\dim \mathfrak{g} = 3$, $|\rho|^2 = \frac{1}{2}$, $h^\vee = 2$; check the strange formula first. P58: for $\gamma = \sqrt{2}(n + \frac{j}{2})$, $|\gamma|^2/2 = (n + \frac{j}{2})^2$, the exponent of q in $\theta_j^{(2)}$; for A_2 , $\langle \omega_1, \omega_1 \rangle = \frac{2}{3}$, $\langle \omega_1, \omega_2 \rangle = \frac{1}{3}$ and $\omega_2 \equiv 2\omega_1 \pmod Q$.

13 Rank two, the alcove, fusion and density (Sections 21–25)

- P59.** (★) List the finite subgroups of $\mathrm{GL}(2, \mathbb{Z})$ up to conjugacy and identify D_2, D_3, D_4, D_6 with $W(A_1 \times A_1)$, $W(A_2)$, $W(B_2)$, $W(G_2)$ by exhibiting the reflections. (Compare P50.)
- P60.** Determine the simplicial automorphism group of the $n \times n$ grid torus with the diagonal $(i, j) - (i+1, j+1)$, modulo translations, and of the triangular-lattice torus. Which is non-abelian? Explain why an automorphism can never realize a Dehn twist, and what the flip passes change so that the twists T_A, T_B and the quarter turn S are read on the square grid.
- P61.** (★) Enumerate P_k^+ for $\mathfrak{su}(2)_k$ and $\mathfrak{su}(3)_k$ and count the states. Show the $\mathfrak{su}(2)$ count is $k+1$ and the $\mathfrak{su}(3)$ count is $(k+1)(k+2)/2$.
- P62.** (★) Starting from Theorem 20.2, derive the explicit $\mathfrak{su}(2)$ character formula of Section 22, keeping track of the level shift $k \mapsto k+2$ and why the denominator has level $2h^\vee = 4$. Then check it numerically at $k=1$: with $w = \frac{1}{2}\langle \alpha, z \rangle$, the quotient $[\theta_{\lambda+1}^{(6)}(w) - \theta_{-(\lambda+1)}^{(6)}(w)] / [\theta_1^{(4)}(w) - \theta_{-1}^{(4)}(w)]$ equals the Frenkel–Kac character $\theta_\lambda^{(2)}(w)/\eta(\tau)$ (P58). Why must the check be made at generic w rather than at $w=0$?
- P63.** Show that setting $W=1$, $\rho=0$, $h^\vee=0$ in Theorem 20.2 recovers Definition 10.1: the abelian register is the no-root case. Why can this substitution not be made in the non-abelian ratio itself?
- P64.** (★) For $\mathfrak{su}(2)_k$ verify that the Kac–Peterson S is symmetric, unitary, and satisfies $S^2 = \text{charge conjugation}$ (here the identity). Compute S and T for $k=1$ and $k=2$. Show that with the $\dim \mathfrak{g}/24$ term in T the relation $(ST)^3 = S^2$ holds exactly, not only projectively.
- P65.** (★) Verify the $\mathfrak{su}(2)_k$ fusion rule of Theorem 23.2 from the Verlinde formula for $k=2$, and reproduce it by Kac–Walton: add the weights of V_μ to λ and reflect into the alcove. Compare with Proposition 19.2: which reflection is new?
- P66.** (★) Prove Theorem 23.4 for $\mathfrak{su}(2)_k$: show $v_\lambda(\sigma) = S_{\lambda\sigma}/S_{0\sigma} = \sin \frac{(\lambda+1)(\sigma+1)\pi}{k+2} / \sin \frac{(\sigma+1)\pi}{k+2}$ equals $\mathrm{ch} V_\lambda$ evaluated at $x = e^{-i\pi(\sigma+1)/(k+2)}$, and verify $v_1(\sigma)^2 = v_0(\sigma) + v_2(\sigma)$ pointwise at $k=2$ for $\sigma=0,1,2$. Explain why the evaluation map is invertible and why this makes fusion “multiplication of values”.
- P67.** In the root coordinates of Section 22.1 with A the (coroot) Gram matrix of A_1 or A_2 : (a) show the characteristic set $P/KQ^\vee$ has $\det(A)K^r$ elements ($2K$ and $3K^2$); (b) show by Poisson summation that

$$\Theta_\mu^{(K)}(-1/\tau, \xi/\tau) = (-i\tau)^{r/2} |P/KQ^\vee|^{-1/2} e^{\pi i K \xi^T A \xi / \tau} \sum_{\mu' \in P/KQ^\vee} e^{-2\pi i \langle \mu, \mu' \rangle / K} \Theta_{\mu'}^{(K)}(\tau, \xi),$$

and deduce that the exponential factors of numerator and denominator in the character ratio combine to $e^{\pi i K \xi^T A \xi / \tau}$; (c) explain why $\Omega = \tau A$ is not a principal period matrix and why $(\tau A) \mapsto -(\tau A)^{-1}$ is not the transformation just derived.

- P68.** (★) Explain, from the two theorems of Section 25, why the torus register is finite for every level and every algebra, and exactly what geometric ingredient (punctures, braids) universality requires. Map each ingredient onto a construction the code already has.
- P69.** State precisely the hypothesis that turns the third claim of Section 1 from algebra into geometry — where the Weyl group acts, and what the surface and the algebra each supply — and design the first experiment that could falsify it. (A torus with D_6 symmetry is a distractor: say why.)

Hints. P59: a finite subgroup preserves a positive definite form, hence is a point group of the square or hexagonal lattice. P62: $\lambda + \rho = \lambda + 1$ in units of the fundamental weight; $|\alpha|^2 = 2$. P64: $\sum_{\mu} \sin \frac{\pi(\lambda+1)(\mu+1)}{k+2} \sin \frac{\pi(\nu+1)(\mu+1)}{k+2}$ is a finite Fourier orthogonality. P66: $\text{ch } V_{\lambda}(x) = (x^{\lambda+1} - x^{-\lambda-1})/(x - x^{-1})$ (P52). P67: Poisson summation in $n \in \mathbb{Z}^r$ for $f(x) = e^{\pi i K \tau x^{\top} A x + 2\pi i K x^{\top} A \xi}$; then group $m \in \mathbb{Z}^r$ into cosets modulo $KA\mathbb{Z}^2$.

The experiment specification

14 Whole harmonic states, pairings and elimination (Sections 26–28)

- P70.** (★) Prove the identifiability proposition of Section 26: with $A = R_{\text{in}}Z$ and $B = R_{\text{out}}Z$, a unique operator T with $B = TA$ defined on every input exists exactly when A is onto and $\ker A \subseteq \ker B$, and then $T = BC$ for any right inverse C of A , with the test $B(I - CA) = 0$. Give a 2×2 example where A is not onto (an input that cannot be prescribed) and one where $\ker A \not\subseteq \ker B$ (an ambiguous output), and say which verifier fixtures (C2) exercise each.
- P71.** For a scalar local system with edge factors g_{ij} define $(d_U f)(v_i v_j) = g_{ij} f(v_j) - f(v_i)$ and $(d_U \alpha)(v_0 v_1 v_2) = g_{01} \alpha(v_1 v_2) - \alpha(v_0 v_2) + \alpha(v_0 v_1)$. Show $d_U^2 f = (g_{01} g_{12} - g_{02}) f(v_2)$ on a triangle, so $d_U^2 = 0$ exactly when the holonomy around every face is trivial (flat), and that the natural pairing of U -cochains with chains needs the dual system U^{-1} for gauge invariance. Explain why the certificates of Section 26 read closedness on cochain images with $\partial^{U^{-1}}$ and coclosedness on the chain frame with ∂^U .
- P72.** Derive the Gram conventions of Section 27. For boundary harmonic chain bases $\Phi, \Phi^{\vee}$ with period matrices $P, P^{\vee}$ of their images and $B_0 = (\Phi^{\vee})^{\top} M_{\text{own}} \Phi$, show that the bilinear pairing in period coordinates is $G_{\text{per}} = (P^{\vee})^{-\top} B_0 P^{-1}$, that the abbreviation $(P^{\vee})^{-\top} P^{-1}$ requires $B_0 = I$, and that with primal/dual transports $M, M^{\vee}$ pairing preservation is $(M^{\vee})^{\top} G_B M = G_A$. Using Remark 7.1, show the pairing is gauge invariant while a primal–primal Gram is not.
- P73.** (★) Prove the elimination identities of Section 28: for $K = \begin{pmatrix} K_{bb} & K_{bz} \\ K_{zb} & K_{zz} \end{pmatrix}$ with K_{zz} invertible, $Kx = 0$ forces $z = -K_{zz}^{-1} K_{zb} b$ and $S_G b = 0$ with $S_G = K_{bb} - K_{bz} K_{zz}^{-1} K_{zb}$. Then minimize $\|u\|^2 + \|r - T_2 u\|^2$ over u and show the minimum is $r^{\dagger} (I + T_2 T_2^{\dagger})^{-1} r$, attained at $u = (I + T_2^{\dagger} T_2)^{-1} T_2^{\dagger} r$. Why is this exact elimination and not a partial trace?

Hints. P70: $c - CAc \in \ker A$ for every c . P73: Woodbury: $I - T_2(I + T_2^{\dagger} T_2)^{-1} T_2^{\dagger} = (I + T_2 T_2^{\dagger})^{-1}$.

15 Quantum norms, channels and acceptance (Sections 29–30)

- P74.** (★) With declared positive $Q_{\text{in}}, Q_{\text{out}}$ and $\hat{T} = Q_{\text{out}}^{1/2} T Q_{\text{in}}^{-1/2}$, show $T^\dagger Q_{\text{out}} T = Q_{\text{in}}$ iff $\hat{T}^\dagger \hat{T} = I$; that the reference marginal of $|\hat{T}\rangle\rangle$ is $(\hat{T}^\dagger \hat{T})^\top / \|\hat{T}\|_F^2$, equal to I/d_{in} for an isometry; and that the Schmidt coefficients of the normalized operator vector are the normalized singular values of $\hat{T}$.
- P75.** For Kraus operators K_a show that $J = \sum_a |K_a\rangle\rangle\langle\langle K_a|$ is positive and that $\text{Tr}_{\text{out}} J = I_{\text{in}}$ iff $\sum_a K_a^\dagger K_a = I$; build the Stinespring isometry $V = \sum_a K_a \otimes |a\rangle_E$ and check $V^\dagger V = I$, $\mathcal{E}(\rho) = \text{Tr}_E(V\rho V^\dagger)$. Then explain by a dimension count why a third boundary component, which enters the relation as a direct summand $B \oplus E$, cannot play the role of the tensor factor $B \otimes E$ of a dilation.
- P76.** Section 29 distinguishes three tensor constructions. (a) Compute $H^2(T^2 \times T^2)$ by Künneth and identify the summand occupied by the degree-one registers of the two factors. (b) Give the dimension of the theta register for a genus- g surface at level k and for two disjoint tori. (c) For $\mathfrak{su}(2)_1$ compare $V_1 \otimes V_1$ with the fusion 1×1 . (d) Two level-two theta factors give four amplitudes; count the amplitudes of a two-qubit gate, of its operator vector, of a direct sum of two two-dimensional port frames, and of a 2×2 selected-pair χ , and explain why matching counts do not define subsystems.
- P77.** V1–V8 require each check to declare whether it compares a geometric computation with an independent oracle or tests a certificate. Classify the verifier’s families R1–R5, H1–H3, J1–J3, C1, D1, U1, T1–T9, F1–F4, G1–G5 and E1–E6 accordingly, naming the oracle or the certified statement.

Hints. P75: use P74 for the marginal of each $|K_a\rangle\rangle$.

16 Neck geometry and encoded-state entropy (Section 33)

- P78.** (★) Prove Proposition 33.2. (a) If the smaller Schmidt probability of a two-qubit state is $p = C\epsilon^2 + O(\epsilon^3)$, show $S = -C\epsilon^2 \log_2(C\epsilon^2) + C\epsilon^2 / \ln 2 + o(\epsilon^2)$, so the entropy is not proportional to ϵ^2 . (b) For $\Omega = \text{diag}(\tau_1, \tau_2) + \epsilon \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ show $\partial_\epsilon \theta_j|_{\epsilon=0} = \theta'_{j_1}(z_1; \tau_1) \theta'_{j_2}(z_2; \tau_2) / (2\pi i k)$, that at $z = 0$ this vanishes for every level-two characteristic, and hence that $p \sim C\epsilon^4$ there. (c) Describe a numerical check of (b) with `ThetaRegister.coherent_state` and `entanglement_entropy`.

Hints. P78: level-two θ_j are even in z : substitute $n \mapsto -n$ ($j = 0$) or $n \mapsto -n - 1$ ($j = 1$).

The measured constructions

17 Reading the records (Section 35)

- P79.** (★) Run the verifier on the collar fixtures,

```
python examples/cobordism/qubit_animation.py verify --tori 2
python examples/cobordism/qubit_animation.py verify --tori 4 \
    --collar-twist swap
```

```
python examples/cobordism/qubit_animation.py theta --tori 4
```

read the records (Section 35.2 names their directories), and for every check name the statement of the reading it measures. Which checks would still pass if the harmonic band were replaced by an arbitrary two-dimensional subspace of closed cochains, and which would fail?

- P80.** (★) The flip passes are declared by their flipped edge, tetrahedron, new edge and lattice map (`DECLARED_FLIP_PASSES`). Pass `b` flips the row edge $(1, 0)$ inside the quadrilateral with vertices $(0, -1)$, $(0, 0)$, $(1, 0)$, $(1, 1)$ to the new edge from $(0, -1)$ to $(1, 1)$; the three lattice maps are

$$L_b = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad L_a = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad L_s = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

(a) Show that the edge set of the flipped grid is the image under L_b of the grid's edge set, the displacements $(1, 0)$, $(0, 1)$ and $(1, 1)$, and that the far marking read through L_b gives $M = L_b^T = T_B$; do the same for L_a and L_s . (b) Show that two `b` passes on the 3×3 grid produce a new edge whose vertex pair is already joined by an edge of the host, while the 5×5 grid separates them. (c) Run

```
python examples/cobordism/qubit_animation.py verify --far-flips b
python examples/cobordism/qubit_animation.py verify --grid 5 --far-flips ab
python examples/cobordism/qubit_animation.py verify --grid 5 --far-flips ba
```

and confirm F2–F4 and the modulus relation F3 of P36.

- P81.** The tube host joins two collars by a prism tube between one far triangle of each. (a) Show that the far boundary is a genus-two surface and that $\chi(\partial W) = -2$, then use $\chi(W) = \frac{1}{2}\chi(\partial W)$ and Mayer–Vietoris to derive the Betti numbers $[1, 4, 2, 0]$ and the restriction rank $4 = \frac{1}{2}b_1(\partial W)$. (b) Explain why the monodromy from the two near tori to Σ_2 is block-diagonal (G2) although the period matrix of Σ_2 has $\Omega_{12} \neq 0$ (G3). (c) With

```
python examples/cobordism/qubit_animation.py theta --tori 4 --join tube
python examples/cobordism/qubit_animation.py theta --tori 4 --join tube \
    --neck-waists 0.25,0.5,1,2,4 --neck-lengths 0.25,0.5,1,2,4
```

record Ω , the coherent-state entropy and the trend of $|\Omega_{12}|$ with the neck, and explain the trend by the plumbing description of a degenerating surface.

- P82.** (★) Build the $\mathbb{Z}/10$ -symmetric genus-two surface,

```
from tessera.quantum import SymmetricGenusTwo, ThetaRegister
import numpy as np
s = SymmetricGenusTwo()
```

and check: 28 vertices, 90 edges, 60 faces, $\chi = -2$; `s.rotation(1)` is a simplicial automorphism (`s.is_automorphism`); `M = s.induced_map(2)` is the matrix of P47 and satisfies $M^T J M = J$, $M^5 = I$. Then fit $\rho_2(M)$ with `ThetaRegister(level=2).weil(M, s.periods.omega)`, verify unitarity and $\rho^5 \propto I$, compute the operator Schmidt rank across the two factors, and count how many of the 36 product stabilizer states are sent to states of exactly one bit of entanglement. Compare with `theta --surface decagon --rotation 2`.

P83. `SurfacePeriods(faces, lengths, a_cycles, b_cycles)` reads a period matrix from a triangulated surface: cotangent weights, harmonic 1-cochains as the kernel of $[d_1; d_0^\top \text{diag}(w)]$, Whitney interpolation, the Hodge star J with $J^2 = -I$, the $-i$ eigenspace and $\Omega = P_B P_A^{-1}$. (a) On a flat square-grid torus of modulus τ check that it returns $\Omega = \tau$ and that the cup pairing gives $A \cdot B = +1$. (b) For the decagon surface read Ω from `s.periods`, check symmetry, positivity of $\text{Im } \Omega$ (eigenvalues 0.46, 1.66), and that Ω is fixed by M under the bridge of P36: $(R + S\Omega)(P + Q\Omega)^{-1} = \Omega$. Why must a surface with an order-five symplectic automorphism have a fixed period matrix, and why does this not make $\Omega = \tau A$ for a root Gram A ?

Hints. P80: the new edge of the second pass has old-coordinate displacement $L_b(1, 2)^\top = (1, 3)^\top$. P81: a 1-handle attached to two tori along disks: $\chi = 0 + 0 - 2$. P82: the product stabilizer states are $|a\rangle \otimes |b\rangle$ with a, b among the six eigenstates of X, Y, Z . P83: an automorphism acts on H^1 by an integer symplectic matrix and on the holomorphic forms by a change of basis.